Hepatic Complications

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49.1 Introduction

The frequency and severity of hepatic complications during hematopoietic cell transplantation (HCT) markedly decreased in the last decade, thanks to improvements in preventive and therapeutic measures for most frequent complications, as fungal infections, hepatotropic viruses, sinusoidal obstruction syndrome/veno-occlusive disease (SOS/VOD), hepatic graft-versus-host disease (GVHD), and improved management of hepatotoxic drugs and iron overload (Hockenbery et al. 2016).

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49.1.1 Classification

The origin of hepatic HCT complications can be:

Multifactorial	**Sinusoidal obstruction syndrome (SOS/VOD)**
Infectious	Bacteria (see Chap. 36), fungi (see Chap. 37), or viruses (CMV, VVZ, and ADV) (see Chap. 38)
Infectious/ immune	**Due to hepatotropic viruses (HBV, HCV, and HEV)**
Immune	Acute or chronic GVHD (see Chaps. 43 and 44) or **autoimmune hepatitis**
Toxic	Due to **hepatotoxic agents** used in HCT
Degenerative	**Cirrhosis and hepatocellular carcinoma**
Iron overload	Hemosiderosis (see Chap. 46)
Ischemic	**Ischemic hepatitis**
Not well established	**Cholangitis lenta, nodular regenerative hyperplasia, focal nodular hyperplasia, and idiopathic hyperammonemia**

In **bold** pathologies analyzed in this chapter

49.2 Veno-Occlusive Disease in Adults

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49.2.1 Introduction

Sinusoidal obstruction syndrome (SOS), also known as veno-occlusive disease (VOD, referred

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to as SOS/VOD hereafter) is a life-threatening complication occurring after HCT (Mohty et al. 2015). Clinical manifestation includes hepatomegaly, hepatalgia, fluid retention with ascites, weight gain, transfusion refractory thrombocytopenia (RT), and jaundice. SOS/VOD usually resolves progressively within a few weeks; nevertheless, in patients with a severe form, the mortality rate is very high (>80%) (Coppell et al. 2010; Richardson et al. 2016). The overall incidence of SOS/VOD in adults can be estimated at around 5–15%, but it varies considerably depending on the presence of risk factors and the conditioning regimen intensity (Mohty et al. 2015; Coppell et al. 2010; Carreras et al. 1998, 2011).

49.2.2 Pathophysiology

Pathophysiology of SOS/VOD is not well known. Conditioning regimens generate toxic metabolites that damage the hepatocytes and activate sinusoidal endothelial cells mainly in zone 3 of the hepatic acinus (Carreras and Diaz-Ricart 2011; Mohty et al. 2016). Activated sinusoidal endothelial cells swell up, leading to the formation of gaps in the sinusoidal barrier. Formed elements of the blood (red blood cells and leukocytes) as well as cellular debris can then pass through these gaps between endothelial cells into the space of Disse and dissect the endothelial lining. This results in a progressive narrowing of the venous lumen, a reduced sinusoidal venous outflow, and ultimately postsinusoidal portal hypertension (Mohty et al. 2015).

Additional information on biomarkers and EASIX index in SOS/VOD could be seen in Sect. 42.1 of the Handbook. EASIX on day 0 seems to be a promising biomarker to identify populations at high risk of SOS/VOD, and studies analyzing correlation with established SOS/VOD risk factors and severity would be important in order to establish how EASIX-d0 can be implemented in routine practice for SOS/VOD diagnosis, severity grading, and treatment initiation (Jiang et al. 2021).

49.2.3 Risk Factors

The accurate definition of SOS/VOD risk factors is indispensable, particularly since they are taken into account in the severity grading. Thus, it is important to classify risk factors as modifiable or unmodifiable to provide some guidance on reducing risk factors and improving patients' management (Table 49.1).

Table 49.1 Unmodifiable and modifiable SOS/VOD risk factors (adults)

Unmodifiable risk factors (in bold the factors with the highest relative risk)
Second HCT
Advanced disease (beyond second CR or relapse)
Primary immunodeficiency diagnosis
Genetic factors (GSTM1 polymorphism, C282Y allele, and *MTHFR* 677CC/1298CC haplotype)
Older patient age
Increased serum transaminase
Karnofsky score below 90%
Metabolic syndrome
Female receiving norethisterone
Previous use of gemtuzumab ozogamicin or inotuzumab ozogamicin
Hepatotoxic drugs
Iron overload (>1.000 ng/mL)
Serum bilirubin > 1.5 mg/L(>26 μmol/L),
Transaminase >2.5 ULN
Preexisting liver disease: hepatic fibrosis, cirrhosis, and active viral hepatitis
Abdominal or hepatic irradiation
Modifiable risk factors and recommendable preventive measures
Conditioning:
High-dose (myeloablative) regimens
Oral or high-dose busulfan
High-dose treosulfan
High-dose TBI-based regimen
Donor:
Unrelated donor
HLA-mismatched donor
GVHD prophylaxis:
Sirolimus + methotrexate + tacrolimus
Methotrexate + cyclosporin or tacrolimus
Non T-cell-depleted transplant
Nutrition:
Use of parenteral nutrition

HCT hematopoietic cell transplantation, *CR* complete remission, *HLA* human leukocyte antigen, *TBI* total body irradiation, *ULN* upper limit of normal, *GVHD* graft-versus-host disease

49.2.4 Diagnosis Criteria

For a very long time, two definitions of SOS/VOD have coexisted, based on the Seattle criteria, reported by McDonald et al. in 1984, and the Baltimore criteria, reported by Jones et al. in 1987. While these definitions were used in clinical practice and research studies, they were not suitable criteria for early diagnosis, and they missed late-onset SOS/VOD. Therefore, in 2016, the European Group for Blood and Marrow Transplantation (EBMT) revised criteria for SOS/VOD were published. Since hyperbilirubinemia and jaundice are almost invariably present in classic SOS/VOD in adult patients, more recently, it was decided to keep the classical original Baltimore criteria for diagnosis of classical SOS/VOD (within 21 days after HCT) in the revised EBMT criteria (Mohty et al. 2016). Indeed, contrary to the Seattle criteria, bilirubin ≥2 mg/dL is mandatory in the Baltimore criteria. In addition, we need to distinguish late-onset SOS/VOD (beyond day 21), where hyperbilirubinemia is less consistent and therefore not mandatory for diagnosis, provided patients present with at least two clinical manifestations (hyperbilirubinemia, painful hepatomegaly, weight gain >5%, and/or ascites) as well as hemodynamic and/or ultrasound evidence of SOS/VOD.

While those criteria have been recently established and no data suggest they should be challenged, we would like to acknowledge that early diagnosis of SOS/VOD can remain difficult in some patients who do not fulfill all SOS/VOD criteria, despite having severe disease. This situation can lead to a delayed initiation of treatment that may have life-threatening consequences. Therefore, the EBMT has updated the previously published criteria, with the addition of a new category of probable SOS/VOD diagnosis (Mohty et al. 2023). Probable SOS/VOD would be defined by two or more of the following five criteria: hyperbilirubinemia, painful hepatomegaly, weight gain >5%, ascites, and/or ultrasound and/or elastography suggestive of SOS/VOD (Table 49.2). SOS/VOD diagnoses based on the previously published EBMT SOS/VOD criteria: association of hyperbilirubinemia with 2 of the following criteria (painful hepatomegaly, weight gain >5%, and/or ascites) will be considered as clinical SOS/VOD, and histologically or hemodynamically proven SOS/VOD will be considered proven SOS/VOD.

Importantly, these criteria overlap with the revised EBMT criteria for late-onset SOS/VOD; therefore, the distinction probable/clinical/proven will also be applied and the only difference for diagnosis between classical and late-onset SOS/VOD will be time of onset (up to day 21 or after day 21).

Diagnostic imaging techniques include hemodynamic, ultrasound, and elastography. Measurement of the hepatic venous pressure gradient (HVPG) through the jugular vein is the most accurate method to confirm the diagnosis of SOS/VOD, since an HVGP >10 mmHg has an extremely high specificity and sensitivity for SOS/VOD diagnosis in patients without previous liver disease

Table 49.2 SOS/VOD criteria for diagnosis (adults)

Probable	Clinical	Proven
Two of the following criteria must be present: – Bilirubin ≥2 mg/dL – Painful hepatomegaly – Weight gain >5% – Ascites – Ultrasound and/or elastography suggestive of SOS/VOD	Bilirubin ≥2 mg/dL and two of the following criteria must be present: – Painful hepatomegaly – Weight gain >5% – Ascites	Histologically proven SOS/VOD or hemodynamically proven (HVPG ≥10 mmHg)
	Onset	
In the first 21 days after HCT: classical SOS/VOD	>21 days after HCT: late-onset SOS/VOD	

For any patient, these symptoms/signs should not be attributable to others causes

(Carreras et al. 1993; Carreras 2015; Shulman et al. 1995; Gressens et al. 2022). However, this technique is invasive, requires experienced staff, and is not routinely available in most centers. Therefore, noninvasive techniques have been developed including ultrasound and elastography. Ultrasound can detect nonspecific abnormalities in SOS/VOD, including hepatomegaly, splenomegaly, gallbladder wall thickening, ascites, and portal venous flow abnormalities (Mahgerefteh et al. 2011; Lassau et al. 1997). A decrease in velocity or reversal of the portal venous flow is considered more specific for SOS/VOD but is inconsistent and usually occurs late in the disease (Mahgerefteh et al. 2011; Lassau et al. 1997; Brown et al. 1990). Importantly, in a study among 106 patients post-allo-HCT, including 10 (9.4%) diagnosed with SOS/VOD, a novel ultrasound scoring, HokUS-10, was established that consisted of 10 parameters (Nishida et al. 2018). The sensitivity and specificity were 100% and 95.8%, respectively. While this score remains to be validated in a larger cohort, it can be useful for ultrasound assessment of SOS/VOD. Of note, there is a direct correlation between the hepatic arterial early acceleration index and HVPG (Tasu et al. 2002), which could be helpful for SOS/VOD diagnosis. Nevertheless, this noninvasive technique requires expertise and is not available routinely.

Liver stiffness measurement (LSM) has been reported as a possible surrogate for portal hypertension and its complications and prompted the evaluation of this technique for the diagnosis of SOS/VOD. Two recent studies have evaluated the impact of LSM in HCT showing excellent specificity and sensitivity to early detect SOS/VOD (Colecchia et al. 2019; Debureaux et al. 2021). Additionally, LSM gradually decreased following successful specific SOS/VOD treatment. LSM can also be evaluated through magnetic resonance imaging (MRI), and increased LSM using magnetic resonance elastography was also reported in patients who developed SOS/VOD after chemotherapy treatment with oxaliplatin, further confirming the role of LSM for SOS/VOD diagnosis (Poker et al. 2022).

Overall, elastography for LSM is sensitive and specific for SOS/VOD diagnosis and is relevant for inclusion in the SOS/VOD diagnostic criteria in addition to hemodynamic and/or ultrasound techniques.

The use of other imaging techniques, including computed tomography (CT) scans and MRI scans, has been investigated in SOS/VOD with no specific findings (Dignan et al. 2013).

49.2.4.1 Severity Grading

According to the EBMT, SOS/VOD is graded in four stages of severity: mild, moderate, severe, and very severe, based on the following parameters: time since the first clinical manifestation of SOS/VOD, bilirubin level and kinetics, transaminase level, weight gain, and renal function (Table 49.3). In the presence of 2 or more

Table 49.3 Severity grading of SOS/VOD in adults

	Mild[a]	Moderate[a]	Severe	Very severe—MOD[b]
Time since clinical symptoms of SOS/VOD	>7 days	5–7 days	≤4 days	Any time
Bilirubin (mg/dL)	≥2 and<3	≥3 and<5	≥5 and<8	≥8
Bilirubin kinetic			Doubling within 48 h	
Transaminases	≤2 × normal	>2 and≤5 × normal	>5 and≤8 × normal	>8 × normal
Weight increase			≥5%	≥10%
Renal function (creatininemia)	Baseline at transplant	<1.5 × baseline at transplant	≥1.5 and<2 × baseline at transplant	≥2 × baseline at transplant or diagnosis of MOD[b]

Patients belong to the category that fulfilled 2 or more criteria. If patients fulfilled 2 or more criteria in 2 different categories, they must be classified in the most severe category between both

[a] In case of presence of 2 or more risk factors for SOS/VOD, patients should be in the upper grade

[b] Patients with multiple organ dysfunction (MOD) must be classified as very severe, and MOD is defined as ≥2 organs from the SOFA score with a score≥2 or an increase ≥2 or organ dysfunction for patients with underlying organ involvement

risk factors, patients are classified in the upper grade. These criteria were validated by Yoon et al. 2019, in a group of 203 patients with SOS/VOD. In these patients, very severe SOS/VOD was associated with a significantly lower OS than the others (58.6% versus 89.3%, $p < 0.0001$) and a higher day +100 transplant-related mortality, being 36.7%, versus 8.3% in mild, 8.0% in moderate, and 2.7% in severe ($p < 0.0001$).

These criteria must be applied once the diagnosis of SOS/VOD is performed according to the revised EBMT diagnosis criteria and can be applied for probable, clinical, or proven SOS/VOD. It is important to evaluate SOS/VOD severity at diagnosis; nevertheless, in some patients SOS/VOD worsens, and we must clearly indicate when we assign SOS/VOD severity grading whether we consider severity at diagnosis or the overall highest severity grade, irrespective of the timing of the grading.

Also, the EBMT clarified the definition of multiple organ dysfunction/multiple organ failure (MOD/MOF). This is particularly important since patients with SOS/VOD who develop MOD/MOF will be classified as very severe (Mohty et al. 2023).

49.2.5 Prophylaxis and Treatment

Regarding SOS/VOD prophylaxis and treatment, we issued recommendations in 2020 (Mohty et al. 2020) that are still accurate today. Defibrotide remains the only agent for the treatment of severe SOS/VOD and should be initiated as soon as possible in those patients. Furthermore, given that early treatment initiation is associated with a higher day +100 OS, and that moderate SOS/VOD is associated with significant mortality (Kernan et al. 2018), we also recommend early initiation of defibrotide in patients with moderate SOS/VOD. For patients with mild SOS/VOD, supportive care must be pursued with close monitoring of severity criteria to allow early initiation of defibrotide in case of worsening. Importantly, defibrotide must be initiated promptly, based on severity criteria as soon as the diagnosis of SOS/VOD is confirmed, irrespective of the diagnostic status (probable, clinical, or proven). Defibrotide is administered at a dose of 25 mg/kg/day for at least 14–21 days and until the resolution of all SOS/VOD symptoms.

Regarding prophylaxis, nonpharmacologic measures to reduce SOS/VOD modifiable risk factors are crucial. For the pharmacologic measures, ursodeoxycholic acid administered from initiation of conditioning until day +90 after transplantation is recommended in adults (Ruutu et al. 2014). Regarding the use of prophylactic defibrotide, a prospective randomized phase III clinical trial compared defibrotide versus best supportive care for prevention of SOS/VOD in 372 pediatric and adult patients at high risk of SOS/VOD after transplantation (NCT02851407) (Grupp et al. 2021). No significant difference was observed between defibrotide and best supportive care groups in the primary end point: SOS/VOD-free survival at day +30 (67% versus 73% respectively, $p = 0.85$). Importantly, there were no differences in adverse events between groups.

49.3 SOS/VOD in Children

Selim Corbacioglu

49.3.1 Introduction

SOS/VOD in children differs in many aspects substantially from adult patients despite a similarity in the underlying pathophysiology. The primary difference is the hepatic immaturity of infants and toddlers. In part, this affects the incidence and the risk factors. There are also predisposing diseases, the clinical presentation, the diagnostic criteria, and finally the indication for prophylaxis that represent major differences in pediatric transplant physicians must be aware of optimizing their approach to one of the most prevalent and deleterious early posttransplant complications in childhood. (Table 49.4 summarizes the major differences between adults and children).

Table 49.4 [a]

Criteria	Children	Adults
Incidence	• Approximately 15–20% • Up to 60% in high-risk patients	• Approximately 5–10%
Risk factors	• Additional pediatric risk factors: – Infants – Pediatric/genetic diseases with incidences above average	• Established risk factors
Clinical presentation	• Late-onset hepatic VOD in 20% • Anicteric hepatic VOD in 30% • Hyperbilirubinemia, if present: – Is frequently preexistent – Occurs late during hepatic SOS/VOD – Hyperbilirubinemia of ≥2 mg/dL is associated with severe hepatic SOS/VOD	• Late-onset hepatic SOS/VOD is rare • Anicteric hepatic SOS/VOD is approximately 15%
Need for proper assessment of ascites and hepatomegaly	• High incidence of disease-related hepatomegaly and ascites pre-HCT	
Diagnostic criteria	• Pediatric EBMT criteria • Inclusion of refractory thrombocytopenia as sensitive marker	• Adult EBMT criteria (obligatory hyperbilirubinemia ≥2 mg/dL) • Baltimore criteria (obligatory hyperbilirubinemia ≥2 mg/dL) • Seattle criteria

[a] Adapted from Corbacioglu S, et al. *Bone Marrow Transplant* 2018; 53:138–45

49.3.2 Incidence

SOS/VOD remains primarily a pediatric disease. The proper incidence is in fact difficult to capture since the diagnostic criteria used impact significantly on this parameter. According to the current literature, a 15% posttransplant incidence (using Seattle/pediatric EBMT criteria (see below) can be assumed (Coppell et al. 2010; Corbacioglu et al. 2012a, b; Xia et al. 2021; Yoon et al.2021), but depending on several influencing factors such as the conditioning regimen and the focus of the respective centers, the incidence can reach 30% and more (Felber et al. 2020). In summary, the overall incidence in children can be considered twice as high as in adults.

49.3.3 Risk Factors

Knowing the conditions that are associated with a high risk for SOS/VOD helps to "earmark" patients for a closer observation during the posttransplant period. The highest risk population are infants and toddlers (<2 years) (Strouse et al. 2018) where SOS/VOD can occur also during conventional chemotherapy, for example, with intensified regimens for infant leukemia. Since risk factors have additive effects (Dalle and Giralt 2016), certain diseases such as malignant infantile osteopetrosis (MIO), congenital macrophage activation syndromes (such as hemophagocytic lymphohistiocytosis - HLH), juvenile myelomonocytic leukemia (JMML), and neuroblastoma, all are highly prevalent in infancy, augment the individual risk significantly (Corbacioglu et al. 2012a, b). Severe iron overload of the liver raises the SOS/VOD risk and is most prevalent in patients in chronic transfusion programs, such as transfusion-dependent thalassemia (TDT), certain patients suffering from sickle cell disease, and aplastic anemia. This risk is more prominent when iron overload leads to an inflammatory response in the liver. In cases where a stagnation of the iron depletion despite an aggressive chelation prior to HCT ("downstaging") is observed, a liver biopsy is highly recommended to assess the underlying inflammatory process. If there is only mild inflammation, the risk for SOS/VOD can be limited despite a high liver iron content. Similarly, patients with an active inflammatory liver disease of an unspecified origin prior to conditioning are at risk. The use of busulfan (inde-

pendent of serum-level measurements) as part of a myeloablative conditioning (MAC) regimen needs to be considered a risk factor, in particular in combination with high-risk diseases such as MIO and HLH (Felber et al. 2020; Strouse et al. 2018). Whereas treosulfan, albeit another alkylator, seems to have a lower detrimental impact on the sinusoidal endothelium with significantly lower incidences of SOS/VOD in high-risk diseases (Shadur et al. 2018; Wustrau et al. 2020).

A prior SOS/VOD and a second MAC-based HCT are additional risk factors. Probably, the group with the highest risk beyond infancy is patients treated with ozogamicin-conjugated monoclonal antibodies preceding a MAC HCT. It is important to mention that even severe SOS/VOD is not limited to patients at risk. In general, awareness of the clinical presentation with or without associated risk factors is pivotal to respond appropriately.

49.3.4 Clinical Presentation

The clinical presentation of SOS/VOD in children also differs in several aspects from adults. The incidence of SOS/VOD peaks around day 12 posttransplant, and 80% of the children are present before day 21. Different from adults, the incidence of anicteric SOS/VOD is approximately 30% (Corbacioglu et al. 2012a, b; Naples et al. 2016) twice as high as in adults, occurring in 80% within the first 21 days post-HCT compared to 50% in adults. Furthermore, the incidence of severe anicteric SOS/VOD with multi-organ dysfunction is higher in children compared to adults (74% versus 59%) (Corbacioglu et al. 2020).

It must be considered that many children present with preexistent hepatomegaly and ascites prior to transplant, prevalent in HLH, OP, TDT, and other diseases frequently transplanted in childhood. Therefore, a pretransplant ultrasound to define the baseline for liver size and free fluid, as required by the pediatric EBMT criteria (pEBMT), is recommended. Right upper quadrant pain, part of the Seattle criteria, is imaginably difficult to assess in infants and was therefore not considered part of the pEBMT criteria.

49.3.5 Diagnostic Criteria

The high incidence of anicteric SOS/VOD discourages the use of criteria that require an obligatory hyperbilirubinemia (>2 mg/dL) as in the Baltimore and adult EBMT criteria (Mohty et al. 2016). In addition, hyperbilirubinemia is a late finding, where even the 2 mg/dL threshold significantly reduces survival and early intervention with Defibrotide has been demonstrated to have a critical impact on morbidity and mortality (Corbacioglu et al. 2020; Kernan et al. 2018). The Seattle criteria, which were used in two large prospective trials (Corbacioglu et al. 2012a, b; Grupp et al. 2023) involving children and adults, omitted the obligatory 2 mg/dL and were until recently more suitable for children.

The pEBMT criteria introduced in 2018 (Table 49.5) intend to cover the pediatric particularities and trigger early therapeutic intervention. With that regard the obligatory bilirubin threshold was omitted reflecting the high incidence of anicteric SOS/VOD. It was replaced by a dynamic approach of rising bilirubin levels on three con-

Table 49.5 [a]

Pediatric EBMT criteria
• No limitation for time of onset of SOS/VOD
• The presence of two or more of the following[b]:
– Otherwise unexplained weight gain on three consecutive days despite the use of diuretics or a weight gain >5% above baseline value within 72 h
– Rising bilirubin from a baseline value on three consecutive days or bilirubin ≥2 mg/dL within 72 h
– Hepatomegaly[c] (best if confirmed by imaging) above the baseline value
– Ascites[c] (best if confirmed by imaging) above the baseline value
– Unexplained consumptive and transfusion refractory thrombocytopenia[d]

CT computed tomography, *MRI* magnetic resonance imaging, *US* ultrasound

[a] Adapted from Corbacioglu S, et al. *Bone Marrow Transplant* 2018; 53: 138–45

[b] With the exclusion of other potential differential diagnoses

[c] Recommended: imaging (US, CT, or MRI) immediately before HCT to determine baseline value for both hepatomegaly and ascites

[d] ≥1 Weight-adjusted platelet transfusion/day to maintain institutional guidelines

secutive days. A similar approach was chosen for weight gain omitting the arbitrary 3% or 5% weight gain of the Seattle and Baltimore criteria, respectively. A baseline abdominal imaging, preferably via ultrasound, is required to verify hepatomegaly and ascites as an individual baseline, particularly important in patients with pre-existing conditions.

The most sensitive criterion added was refractory thrombocytopenia. SOS/VOD is a sinusoidal disease, with a consumptive and transfusion refractory thrombocytopenia (RT) being the earliest symptom. Therefore, RT is the most sensitive trigger for intervention. RT has been described by several authors as an early sign and, if prolonged as a predictor of poor outcome, including McDonalds, was never introduced as a criterion (McDonald et al. 1993; Embaby et al. 2020; Roeker et al. 2019). Several publications confirmed the validity of RT as an early trigger by comparing pEBMT with the established criteria. Szmit et al. (2020) showed in a single center that using modified Seattle criteria delayed diagnosis by up to 11 days (median 3 days) compared to pEBMT criteria. Another single-center analysis in 226 pediatric patients found a shortened time-to-diagnosis also by 3 days, with a 75% incidence of RT as the first symptom (Ragoonanan et al. 2021). With regards to scientific objectives, it is pivotal to be aware of the impact of the different diagnostic criteria on the incidence of SOS/VOD, with a doubling or even quadrupling the incidence by using Baltimore versus Seattle criteria (Coppell et al. 2010; Yakushijin et al. 2005) and another rise of approximately 5% from to Seattle to pEBMT criteria.

49.3.6 Therapeutic Intervention

The therapeutic interventions in children differ only in parts from adults, with defibrotide being the only licensed drug to specifically treat SOS/VOD. Early intervention demonstrated to have a pivotal impact on morbidity and mortality. A bilirubin level above 2 mg/dL in all ages is a predictor of a poor day 100 survival (Corbacioglu et al. 2020). Earlier initiation with defibrotide was associated with higher day 100 survival in the subgroup of patients with multi-organ dysfunction (Kernan et al. 2018).

When comparing Seattle criteria-triggered intervention with defibrotide with pEBMT criteria a significantly better overall survival, a five times lower transplant-related mortality and a shortened length of hospitalization in the median of 12 days was observed (Szmit et al. 2020). Next to the established supportive measures of balanced fluid management to avoid prerenal failure, avoidance of hepatotoxic drugs, and careful management of hemostasis, a particular recommendation for toddlers is to consider early paracentesis for ascitic drainage to avoid pulmonary impairment/insufficiency due to infradiaphragmatic expansion into the thoracic cavity.

49.3.7 Prophylaxis

Regarding the efficacy of defibrotide for prophylaxis, the body of evidence seems to be contradictory. Next to several retrospective analyses and case series, the pediatric prevention trial (NCT00272948) was a prospective randomized trial conducted in 360 high-risk pediatric patients that demonstrated efficacy in the prevention of SOS/VOD by reducing the incidence from 20 to 12% ($p = 0.049$; per protocol population: $p = 0.022$) (Corbacioglu et al. 2012a). On the other hand, the Harmony Trial (NCT02851407), another prospective randomized trial with a similar sample size, that also included 50% adults, reported that defibrotide was not effective for the prevention of SOS/VOD. This discrepancy can be solved when the latter trial is scrutinized regarding sample size calculation, end points, and general design (Corbacioglu et al. 2023). Therefore, several aspects encourage a critical discussion on the need for prophylactic intervention with defibrotide in children. The disease is very prevalent, and a delayed intervention can affect morbidity and mortality significantly. The diagnostic criteria remain based on clinical observation only and might under certain circumstances lead to a substantial therapeutic delay. On

the other hand, sensitized diagnostic criteria such as the pEBMT criteria can trigger early/preemptive therapeutic intervention with defibrotide by several days. In conclusion, the question remains if the indication for prophylactic intervention with defibrotide is obsolete or remains an unmet need in children to cover this gap in a high-risk pediatric population.

49.4 Hepatotropic Viruses

Rafael de la Cámara

49.4.1 Hepatitis A Virus (HAV)

Hepatitis A virus (HAV) is a non-enveloped hepatotropic virus, now named *Hepatovirus A*, classified in the genus *Hepatovirus* within the family *Picornaviridae*.

Most countries of the European Economic Area (EEA) currently experience very low or low HAV endemicity.

HAV is primarily transmitted via the fecal–oral route and is acquired through ingestion of contaminated food or water, sexual, or another direct contact with an infected individual. On rare occasions, HAV has been transmitted by transfusion of blood or blood products collected from donors during the viremic phase of the infection. Blood products and HCT donors are not routinely tested for HAV.

49.4.1.1 Clinical Symptoms

Hepatitis due to HAV is usually mild and self-limited when healthy persons are infected. Disease severity increases in older or immunocompromised, have chronic liver disease, or have other underlying health conditions. No chronic infection is known to occur, although prolonged, relapsing hepatitis occurs in 15% of cases. Very severe disease is unusual. There is no specific series of HAV hepatitis in HCT patients.

49.4.1.2 Diagnostics

Anti-HAV IgM establishes the diagnosis of acute hepatitis A and can remain elevated for 3–12 months following infection. Polymerase chain reaction (PCR) is the preferred method in the HCT setting. Anti-HAV IgG generally persists for the duration of a patient's life following infection or vaccination.

Liver function tests (LFTs) should be performed in donors before HCT harvesting. Donors with abnormal LFT should be tested for anti-HAV Immunoglobulin (Ig) M. If HAV is detected, the donation should be delayed until HAV-RNA is no longer detectable in the donor.

49.4.1.3 Prevention

HCT is not recommended if the donor or the recipient is viremic for HAV because of an increased risk of SOS/VOD (Nelson et al. 2020).

Vaccination should be considered in HAV-IgG-negative patients at risk. There is limited experience of HAV vaccination in HCT. Limited data suggest that doubling of the standard antigen might increase response rates.

In some countries, there are specific HAV Ig products that can be used for both preexposure for travelers to intermediate or high-endemicity areas prior to travel and postexposure prophylaxis of HAV (within 2 weeks of the exposure).

Persons with chronic liver disease have a higher likelihood of fulminant hepatitis if not previously vaccinated or immune against HAV. HCT with chronic liver disease should be tested for antibodies against HAV and, if negative, should be advised to be vaccinated against HAV.

49.4.1.4 Treatment

Symptomatic (Deasy and Kim 2020).

49.4.2 Hepatitis B Virus (HBV)

HBV is a double-stranded DNA virus, a species of the genus *Orthohepadnavirus*, within the family *Hepadnaviridae.*

HBV is a frequent infection worldwide, and consequently, HBV infection in HCT candidates is a frequent situation. An infection due to HBV does not preclude an HCT, but the liver disease caused by this virus can.

49.4.2.1 Clinical Symptoms

After primary infection, even in the case of HBsAg seroconversion (became negative with positive anti-HBs), HBV probably persists lifelong in the nucleus of hepatocytes and, therefore, can reactivate after treatment-induced loss of immune control as is the case of HCT. Hepatitis, including cases of fulminant hepatic failure, typically occurs after immune system reconstitution, de novo recognition, and destruction of HBV-infected hepatocytes. Reactivation of resolved HBV infection, known as reverse seroconversion (RS), is an important and late complication of HCT, with a median of 18–20 months after transplant. The risk of reactivation is higher among those with HBsAg (+) than in those with HBsAg (−) and anti-HBc (+), in those with low anti-HBs (<10 IU/L), extensive chronic GVHD, and higher in allogeneic compared to auto-HCT. Fibrosing cholestatic hepatitis can be a consequence of HBV reactivation. Nonetheless, HBV infection does not affect 10-year survival.

49.4.2.2 Diagnostics

All donors and recipients must be screened for HBsAg, anti-HBc, anti-HBs, and HBV-DNA (or at least if anti-HBc or HBsAg are detected). The combination of serology, DNA-HBV, and level of transaminases classified the different types of HBV infections (Table 49.6).

Table 49.6 Different types of HBV infection

Not infected, not vaccinated	**Immunized by vaccination**
HBsAg (−) Anti-HBs (−) Anti-HBc (−) DNA-HBV (−)	HBsAg (−) Anti-HBs (+) Anti-HBc (−) DNA-HBV (−)
Acute infection, window period	**Acute infection**
HBsAg (−) Anti-HBs (−) IgM anti-HBc (+) DNA-HBV (±)	HBsAg (+) Anti-HBs (−) IgM anti-HBc (+) DNA-HBV (+)
Acute hepatitis in resolution	**Immunized after infection (occult HBV infection)**
HBsAg (±) Anti-HBs (±) IgM anti-HBc (−) IgG anti-HBc (+) DNA-HBV (±)	HBsAg (−) Anti-HBs (+) IgM anti-HBc (−) IgG anti-HBc (+) DNA-HBV (−)
Chronic active hepatitis[a]	**Chronic carrier**
HBsAg (high +) Anti-HBs (−) IgM anti-HBc (−) IgG anti-HBc (+) HBeAg (+), anti-HBe (−) DNA-HBV (+) (>2000 UI/mL) Elevated transaminases	HBsAg (low +) Anti-HBs (−) IgM anti-HBc (−) IgG anti-HBc (+) HBeAg (−), anti-HBe (+) DNA-HBV (+) (<2000 UI/mL) normal transaminases

Adapted from Carreras E, Rovira M. Valcarcel D (eds) Manual de Trasplante Hematopoyético, 2022. Chap. 4.18

[a] Chronic hepatitis B can be HBeAg-negative, usually with detectable anti-HBe, and persistent or fluctuating moderate to high levels of serum HBV DNA (often lower than in HBeAg-positive patients). Most of these subjects harbor HBV variants in the precore and/or the basal core promoter regions that impair or abolish HBeAg expression

49.4.2.3 Prevention

The most important practical issue with HBV in HCT is the prevention of severe or fatal hepatitis due to HBV (Siyahian et al. 2018).

(a) DNA-HBV or HBsAg-positive patients

Antiviral prophylaxis should be given to all DNA-HBV or HBsAg-positive patients. Tenofovir (limited experience in HCT patients) or entecavir are the drugs of choice, starting before the conditioning regimen and maintained at least for 1 year after HCT, and longer if GVHD is present or immunosuppression is given. Lamivudine is not recommended. After stopping antiviral prophylaxis, HBV monitoring every 3 months is recommended as late reactivation with fulminant hepatitis has been described.

(b) Anti-HBc-positive but HBsAg/DNA-HBV-negative patients

Two strategies can be used for the prevention of HBV hepatitis in these patients: prophylaxis with antivirals or a preemptive therapy with DNA-HBV monitoring.

The ECIL recommends antiviral prophylaxis (with tenofovir or entecavir) to all anti-HBc-positive patients who undergo a HCT (Mallet et al. 2016).

The European Association for the Study of the Liver recommends antiviral prophylaxis when the risk of HBV reactivation is

high (>10%) (EASL 2017). In these patients, lamivudine is also an option. For moderate- (<10%) or low-risk (1%) reactivation patients, preemptive therapy, not prophylaxis, is generally recommended. Preemptive therapy is based on monitoring HBsAg and HBV DNA every 1–3 months and starting antiviral treatment as early as HBV DNA is detectable or HBsAg seroconversion, independently of alanine transaminase (ALT) levels.

(c) HBsAg (or DNA-HBV)-positive donors

An HBsAg (+) donor transmits HBV to the recipient in 22–44% of cases. Antiviral therapy of the donor and, if possible, antiviral prophylaxis of the recipient are recommended. The addition of hepatitis B immune globulin can be considered. Pre-HCT vaccination of the recipient, if possible, could decrease the transmission of HBV from the donor.

49.4.2.4 Vaccination

Vaccination of anti-HBc-negative and anti-HBs-negative patients after HCT is recommended. Double vaccine doses may be required to achieve an anti-HBs response in immunocompromised patients (0–1–2–6 months). Vaccination can reduce the risk of reverse seroconversion in anti-HBc-positive patients (Hammond et al. 2022).

The transfer of HBV immunity from donor to recipient has been described. Vaccination of the donor should be done only in the interest of the donor.

Patients with HBV infection should be tested for antibodies against HAV and, if negative for anti-HAV, should be advised to be vaccinated against HAV.

49.4.2.5 Treatment

In HCT candidates with chronic hepatitis based on biopsy or positive HBsAg or high levels of HBV DNA, transplant procedure should be delayed when possible, and antiviral therapy should be given for 3–6 months before conditioning.

Treatment of patients with acute or chronic HBV hepatitis should be done in collaboration with hepatology. Acute hepatitis after HCT should also be treated to prevent a bad or fatal course.

49.4.3 Hepatitis C Virus (HCV)

The hepatitis C virus (HCV), now named *Hepacivirus C*, is a small, enveloped, single-stranded, RNA virus, the member of the genus *Hepacivirus* in the family *Flaviviridae*. HCV is a frequent infection worldwide, and consequently, HCV infection in HCT candidates is a frequent situation. The hepatitis C virus is a bloodborne virus. Transmission of HCV by blood transfusion is an exceptional event in developed countries. In HCT, the most important aspect is the detection of HCV infection in order to implement treatment to avoid liver damage.

49.4.3.1 Clinical Symptoms

HCV can cause both acute and chronic hepatitis infection. The majority of HCV infections are asymptomatic.

An infection due to HCV does not preclude an HCT, but the possible existence of hepatic fibrosis or the presence of cirrhosis and hypertension in HCV-RNA (+) patients should be evaluated prior to the transplant. Liver fibrosis is a risk factor for SOS/VOD and drug toxicity. Cirrhosis and a worse outcome have been documented after HCT. Nonetheless, HCV infection does not affect the 10-year survival.

49.4.3.2 Diagnostics

All donors and recipients must be screened for HCV before transplant by serology and HCV-RNA PCR. In cases of spontaneous or treatment-induced viral elimination, anti-HCV antibodies persist for life in the absence of HCV RNA. Patients who have successfully eliminated HCV in the past are not at risk of reactivation under immunosuppressive therapy, but reinfection with HCV is possible.

Close monitoring of LFT and HCV-RNA is recommended in infected patients.

49.4.3.3 Prevention

There is no vaccine for HCV, so prevention of infection depends on reducing the risk of exposure to the virus in healthcare settings and avoiding personal practices that are associated with the risk of transmission (unsafe sex, injecting drugs, tattooing, acupuncture, and body piercing).

SOS-sparing regimens should be considered in HCV RNA-positive patients with significant liver fibrosis.

The presence of HCV-RNA positive in the recipient does not constitute a contraindication for HCT, but antiviral therapy should be considered, if possible, to postpone the HCT to allow completion of a treatment course.

If the donor is HCV-RNA-positive, the infection will be transmitted to the recipient in all cases. An HCV-RNA-positive donor could be considered if other donor options are considered inferior. In this case, the donor should be rapidly evaluated by a hepatologist, and treatment with direct-acting antivirals (DAAs) should be considered.

49.4.3.4 Treatment

The treatment of HCV has entirely changed with the incorporation of direct-acting antivirals (DAAs), which obtain cure in over 90% of patients. Antiviral treatment should be considered for all HCV-RNA-positive hematological patients. HCV can be treated concomitantly with chemotherapy if treatment of hematological malignancy is urgent. This should be done in consultation with an expert hepatologist. If possible, all HCV-RNA-positive patients should be treated before the transplant.

49.4.4 Hepatitis E Virus (HEV)

Hepatitis E is caused by the HEV, a positive-stranded RNA virus of the *Hepeviridae* family, genus *Orthohepevirus A*.

HEV is divided into 4 genotypes. Genotypes 1 and 2 are more virulent and infect humans only, while genotypes 3 and 4 are zoonotic. In Europe, autochthonous infections are mostly related to HEV-3. The seroprevalence of HEV in Europe varies between and within countries (1 to >50%).

The main source of spread varies between different parts of the world. Hepatitis E is a waterborne infection caused by HEV genotype 1 or 2 in developing countries, while in developed countries, autochthonous hepatitis E is a zoonotic infection (reservoir in pigs or wild boar), caused by HEV genotypes 3 and 4. There is no evidence of sexual transmission of HEV. Transfusion (plasma, platelet concentrates, and red blood cell concentrates) or transplantation-transmitted HEV infections have been observed sporadically. In Europe, where HEV-3 is endemic, the infection is not associated with severe disease in pregnant women, and thus, they are not considered as a risk group (Mikulska et al. 2022).

Patients undergoing HCT might be at risk of acquiring HEV through blood transfusions. There is a possibility for HEV transfer from stem cell donors. Currently, there is no possibility of calculating the risk–benefit ratio of systematically testing donors for HEV RNA.

49.4.4.1 Clinical Symptoms

HEV can cause acute or chronic hepatitis. Acute infections cause self-limiting hepatitis but can become chronic in immunocompromised patients, like HCT patients, with the risk of the development of severe liver cirrhosis. In a few cases, the acute infection can result in fulminant hepatitis with acute liver failure. No case of fulminant HEV in HCT recipients has been described so far. Patients with preexisting chronic liver disease are at risk of severe disease progression with liver failure. Several extrahepatic manifestations have been described associated with HEV. The probably more important clinical picture in HCT recipients is chronic hepatitis, usually showing limited symptoms of hepatitis or nonspecific clinical symptoms, since rapid progression to cirrhosis has been reported in IS patients.

49.4.4.2 Diagnostics

Nucleic Acid Amplification Testing (NAT) testing should be preferred over serology to diagnose HEV infection, as serological tests vary in sensitivity and specificity. In HCT, no routine screening for antibodies against HEV or RNA-HEV is done either in the donor or in the recipient.

49.4.4.3 Prevention

So far, the detection of HEV-RNA neither in the donor nor in the recipient can be considered as an absolute contraindication for HCT. HCT with an HEV-RNA-positive donor could be considered if other donor options are considered inferior. In this case, treatment with Ribavirin (RBV) of the recipient could be considered.

HCT recipients should be informed about the risks of foodborne HEV transmission by avoiding the consumption of undercooked meat. When traveling to countries with poor sanitation, it is advisable to boil water for drinking and for brushing teeth.

A vaccine has been developed a licensed in China (2011) but is not licensed in Europe and the USA or recommended for use by WHO.

49.4.4.4 Treatment

For patients with HEV infection, a decrease in the dose of immunosuppressive drugs could be considered. Ribavirin has been suggested as a treatment for chronic infection based on case reports and small case series. However, no controlled data exist.

49.5 Other Hepatic Complications

Enric Carreras and Tapani Ruutu

49.5.1 Autoimmune Hepatitis

The main problem with these hepatitis is how to differentiate them from hepatic GVHD, as the pathogenesis, clinical manifestations, and biological changes are virtually identical (Dalekos et al. 2002; Ruutu and Carreras 2019) as shown in the table:

Variables	Autoimmune hepatitis	Hepatic GVHD
Jaundice	Generally mild	Various degrees
Other symptoms	Fatigue and malaise, but often asymptomatic	Hepatic tenderness, choluria, acholia, and anorexia, **almost always GVHD in other organs**
Increased AST	Moderate to severe	Minimal
Increased GGT	Marked	Normal or decreased
Type of auto-antibodies	Characteristic of AIH, Ab type 2 (**ALKM and ALC-1**)	Similar Ab but usually of AIH type 1 (ANA and ANCA)
Pathology	Inflammatory infiltrate in the portal area, often penetrating into lobules	Inflammatory infiltrate, **bile duct leakage, degeneration of the ductal epithelium**, and cholestasis
Cirrhosis	May be present	Exceptional
Steroid resp.	Excellent	Variable according to severity

In **bold** main differential data

Ab autoantibodies, *AIH* autoimmune hepatitis, *ANA* antinuclear antibodies, *ANCA* anti-cytoplasm antibodies, *ALKM* microsomal Ab. against liver and kidney, *AC1* anti-cytosol type 1

49.5.2 Drug-Induced Hepatitis

Most relevant agents[a]	Presentation
Triazole antifungals[b]	Cholestatic[c] or hepatocellular[d] hepatitis, hepatic insuf.
Echinocandins	Mild-moderate or hepatocellular hepatitis
Fluoroquinolones	Hepatocellular hepatitis
TMP-SMX	Hepatocellular hepatitis
Rapamycin	Hepatocellular damage and increased risk of SOS
Anticonvulsants	Hepatocellular hepatitis or cholestatic hepatitis
NSAIDs	Hepatocellular or cholestatic hepatitis
Paracetamol	Hepatocellular or cholestatic hepatitis
Ranitidine	Cholestatic hepatitis and eosinophilic infiltration
Amoxicillin-clavulanic acid	Cholestatic and/or hepatocellular hepatitis

[a] There are online resources for toxicity and interaction queries such as LiverTox: www.ncbi.nlm.nih.gov/books/

[b] Voriconazole and posaconazole

[c] Liver damage with predominant elevation of transaminases

[d] Liver damage with predominant elevation of cholestasis and bilirubin enzymes

49.5.3 Cirrhosis

Based on historical data, it can be said that evolution to cirrhosis has been described in the following cases:

- **HBV**: progression to cirrhosis after HCT is exceptional.
- **HCV**: a case–control study evaluated the prevalence of cirrhosis in 3721 long-term survivors of HCT. In patients who survived ≥1 year after HCT, cirrhosis (clinical or histological) was detected in 31 of the 3721 patients (1%). The median time from HCT to cirrhosis was 10 years. HCV was the etiology of cirrhosis in 25 of the 31 patients, compared to 14 of the 31 control patients ($p = 0.01$).

In a single-center retrospective study, 96 HCV-infected patients during the HCT period were compared with a control group of 158 HCV patients without HCT ($p = 0.01$). The cumulative incidence of cirrhosis at 20 years of HCT was 24%, and the median time to cirrhosis was 18 years, compared to 40 years in the control group (Peffault De Latour et al. 2004).
In another study, the cumulative incidence of severe liver complications in HCV-infected patients was 11.7% at 20 years in a multicenter cohort (Ljungman et al. 2012).

- **HEV**: the incidence of progression to cirrhosis of HCTs with HEV infection is unknown, but isolated rapidly progressive cases have been reported (Swartling et al. 2020).

Poorly compensated cirrhosis is a formal contraindication for HCT due to the very high risk of developing SOS after MAC (Swartling et al. 2020). Even compensated cirrhosis has a high likelihood of hepatic decompensation even on reduced-intensity conditioning (RIC) or non-myeloablative conditioning (NMC) (Hogan et al. 2004). It is not known whether these historical data remain true after the availability of the new antiviral agents.

49.5.4 Hepatocellular Carcinoma (HCC)

Hepatocellular carcinoma (HCC) is the most common liver cancer and is the fourth leading cause of cancer death. The most important risk factor for developing HCC is cirrhosis. Therefore, chronic liver disease due to HBV or HCV and hepatic steatosis play a key role in the pathogenesis of HCC (reviewed in Gilman et al. 2021).

The cumulative incidence of HCC after HCT is 5%, 20 years after the procedure (Peffault De Latour et al. 2004). Therefore, at-risk patients should undergo surveillance with liver ultrasound scans every 6–12 m, according to international guidelines.

Patients may be asymptomatic or have clinical manifestations of cirrhosis or decompensated liver disease.

Early diagnosis can be made by periodic abdominal ultrasound, with or without levels of alpha-fetoprotein (AFP) levels. If a lesion is detected on ultrasound, imaging should be completed with CT or MRI. Definitive diagnosis should be established by liver biopsy, despite an elevated AFP is practically confirmatory of the diagnosis.

Therapeutic options may be curative or noncurative. Curative options include surgical resection, ablation, and liver transplantation. Noncurative options include embolization, radiation, and systemic chemotherapy.

49.5.5 Nodular Regenerative Hyperplasia (NRH)

Nodular regenerative hyperplasia (NRH) is a form of noncirrhotic portal hypertension with small regenerative nodules in the liver (reviewed in Gilman et al. 2021). After HCT, it is occasionally seen in patients with previous SOS. Its pathogenesis is probably the consequence of changes in hepatic blood flow with atrophy of zone 3 of the acinus and hypertrophy of zone 1 (without fibrosis).

The diagnosis may be suspected in the presence of a silent course (except for thrombocytopenia and increased alkaline phosphatase) toward portal hypertension (variceal bleeding, ascites, and hepato-splenomegaly). MRI shows characteristic nodular images. Liver biopsy allows to establish the diagnosis and rule out carcinoma or cirrhosis. This study should not be performed by transjugular or fine needle biopsy because these procedures do not provide sufficient diagnostic material (McDonald 2010). Management includes treatment of the underlying disorder and management of portal hypertension.

49.5.6 Focal Nodular Hyperplasia (FNH)

Focal nodular hyperplasia (FNH) is a benign liver lesion being the second most common hepatic tumor (reviewed in Gilman et al. 2021). Its pathogenesis is unknown. In HCT, it is thought to be a consequence of sinusoidal injury from the conditioning regimen or by agents that subsequently cause endothelial damage. Patients with FNH are usually asymptomatic.

Its incidence was assessed in a prospective study of 138 HCT (70% children and 30% adults) who were being followed up for screening for early detection of secondary hemochromatosis. The diagnosis of FNH was made by MRI in 16 (12%) with a median HCT diagnosis time of 6.4 years.

In a retrospective study of 87 pediatric patients after HCT, the diagnosis of FNH was established in 10 patients (11%) with a median time to diagnosis of 7 years. There was no malignant transformation. In a third study with 324 HCT, 17 patients (5.2%) with FNH were identified after a median time of 5.7 years. There was no malignant transformation.

The diagnosis is usually made incidentally by performing an ultrasound scan that detects an isoechoic lesion. The confirmatory diagnosis includes CT or MRI, which may detect a characteristic central scar. On CT, the lesions appear diffuse, homogeneous and hyperdense, reflecting the central scar. On MRI, the lesions have a hypervascular (arterial) appearance, an isointense (portal venous) or hypointense signal on T1 sequences (without contrast), and an isointense or hyperintense signal on T2 sequences (without contrast). Liver biopsy is not routinely indicated to confirm the diagnosis. Treatment should be conservative, and follow-up imaging tests are not needed.

49.5.7 Idiopathic Hyperammonemia

A rare but lethal syndrome with hyperammonemia and coma after conditioning for HCT has been described. Patients present with progressive lethargy, confusion, weakness, incoordination, vomiting, hyperventilation with respiratory alkalosis, and plasma ammonia >200 μmol/L. The pathogenesis of idiopathic hyperammonemia likely involves a latent genetic disorder similar to ornithine transcarbamylase deficiency (McDonald 2010).

49.5.8 Cholangitis Lenta (CL)

CL is a hyperbilirubinemia frequently observed in neutropenic and febrile patients with intestinal mucosal lesions due to the conditioning regimen. Hepatocyte retention of conjugated bilirubin appears to be mediated by endotoxins, IL6 and TNFα. The exact pathophysiologic mechanisms of sluggish cholangitis remain unclear. It is hypothesized that bacterial endotoxin release and subsequent damage may cause ductal proliferation and impaired bile flow (Torous et al. 2017). Although this disorder is often referred to as "cholestasis of sepsis," it can be seen in patients with isolated fever without focality, or in patients with a localized infection in the lungs and soft tissues (McDonald 2010).

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